

AI-ML ENABLED · PROOF OF CONCEPT

Digital Twin Implementation at JNPA

Bidder Briefing — Pilot Use Cases, Data Sharing & Evaluation

16 JULY 2026

09:30 – 17:30 · JNPA

3 PILOT USE CASES

DATA SHARING PLATFORM

EVALUATION APPROACH

BIDDER BRIEFING

Jawaharlal Nehru Port Authority

Tender Ref: GEM / 2026 / B / 7297343

Schedule for the Day

09:30 – 17:30

1	09:30 – 10:00	30 min	Bidder Registration	
2	10:00 – 10:20	20 min	Dy. Chairman's Address	
3	10:30 – 11:15	45 min	Use Case I - Vessel Traffic Management & Optimization	Deputy Conservator
4	11:15 – 12:00	45 min	Use Case II- Improved Cargo Handling & Logistics Optimization	Chief General Manager (Traffic)
5	12:00 – 12:45	45 min	Use Case III- Traffic Monitoring & Vehicular Decongestion	Chief General Manager (Traffic)
6	12:45 – 13:30	45 min	Interaction with Bidders	
	13:30 – 14:30	1 hr	Lunch Break	
7	14:30 – 16:00	1 hr 30 min	Data Sharing Platform & Process	PMC
	16:00 – 16:30	30 min	Tea & Snacks	
8	16:30 – 17:15	45 min	Discussion & Q&A with Bidders	
9	17:15 – 17:30	15 min	Closing Remarks	

● Pilot use cases ● Data sharing ● Bidder interaction

Interaction with Bidders

Digital Twin – The JNPA Perspective

Four levels of digital twin abstraction — JNPA's deployment is a Process Level Digital Twin envisioned to help each stakeholder



What Each Use Case may cover

	UC I Vessel & Marine	UC II Cargo Handling	UC III Traffic & Decongestion
Solution Approach & Edge Cases <i>Concepts, activity diagram, flow chart. Edge cases: shutout, scanning list, roll-over.</i>	- Vessel bunching - Weather (rain) - Tidal · dredging - Pilot / tug availability - Oil spill, accident	- Shutout · scanning - Equipment availability - Yard congestion - Hazardous cargo - ITRHO / ITT	- Driver shortage - Accident — premises / enroute - Transporter blacklist - ECY congestion - Illegal parking
AI / ML Tools <i>Applied across the use cases.</i>	- DUKC - Extend tidal window - ETA / ETD → berth utilisation - Port craft assignment - Predicting turnaround time	- Discharge rate - Rake assignment - Lane planning - Pendency optimisation & alerts	- Parking guidance - ETA estimate - TT double trip - RMS · TAS slots - Camera AI (Object detection, Classification, Identification- Trailer, Container, Driver, Document OCR)
Data Integration & Fallback <i>Min. 1 API + 1 fallback per use case. State data volumes.</i>	- Global visibility (VTS) - Weather data - Satellite maps (BhooNidhi) / ocean — offline copy - Channel depth — static overlay - Berthing data (VMS)	- LDB — daily berthing report - FOIS · ULIP - iWebs (email) - NLP-M - ICEGATE	- NVR — network video recorder - FASTag - PDP - LDB - Fleet tracking data
Dashboard & KPI	- JIT - Pre-berthing / pre-sailing delay - Avg. vessel TAT - Port craft optimisation - Actual vs predicted	- Rake TAT · ITT TAT - Trailer TAT · scanner TAT - Buffer pendency - Mixed train - Trans-shipment TAT	- Gate queue wait time - Avg. gate txn time - ECY TRT - TAT inside port
What-if & Reactive Twin	- Extended berth window - Tidal window missed by 30 min — cascade - Vessel bunching - Weather — wind > 30 knots - Reactive: tidal + port craft + vessels at sea → DUKC → confidence	- Rail side · road side - Customs - Equipment availability - Impex bunching - Empty discharge & load impact on gates	- ECY & gate congestion 3 main road bottlenecks - Heavy rainfall · driver / fuel shortage - Reefer availability at CPP - Reactive: guidance on weather, routes & TAS

Expected Data to be Submitted — by Workstream

Expected Data to be Submitted · one submission per use case, per workstream	
Solution Approach & Edge Cases	Diagrams: Activity Diagram · Flow Charts Concepts (in-in flow) → person activity — covering the micro / edge cases for each use case.
AI / ML Tools	Tabular description with columns: Use Case Solved · Training Data (Features) · Objective Function · Model Used · Rationale · Link to Model Weights · Validation Data · Accuracy
Data Integration & Fallback	Tabular description with columns: API Integration · Rationale & Usage · Fallback Mechanism · Credentials · Consumption Mechanism
Dashboard & KPI	Tabular description with columns: KPI Measure · Methodology for Calculation · Input Data · Actionable Insights Plus images and videos of the dashboard.
What-if & Reactive Twin	Response to the what-if scenarios shared in the final week The team to show how the system responded, and the reactive response it provided.

Formats in scope: .json · .xml · .txt · .pdf · .jpg / imagery · email — the five workstreams are to be submitted for each of the three use cases.

PoC - EVALUATION STRATEGY & ACCEPTANCE METHODOLOGY

01Functional Example: Marine operations Yard operations Gate operations Rail operations	02AI Example: Object, vehicle, container, driver detection , OCR for documents Tracking continuity for multiple cameras, Event classification	03Live Operations Example: Continuous run on live data - Validated vs API	04Integration Example: VTMS · PCS · ERP · GIS, CCTV · RFID · AIS · IoT	05Workflow Example : Event → detection → alert → notification → acknowledgement → assignment → resolution → closure → audit trail
06Performance Example: Dashboard refresh Alert generation API response AI inference GIS refresh	07Cybersecurity Example: RBAC · encryption API auth · audit logs Event logging	08Stress Testing Example: Peak vessel & truck arrivals · camera streams High load, Network latency	09Failover & Exceptions Example: Camera · network · API · Database · server Edge system offline	10Evidence Example: Machine & AI inference, logs · database records Operational records Screenshots · video

Accessing the Data, Raising Queries & Submitting

All via Next Cloud

01 How the data is accessed

- Access is through Next Cloud
- Accessed via a whitelisted IP & registered email ID — **both to be shared by the bidder by return email**
- Sample data is stored in the provided folder
- Data provisioned in subsequent weeks may also come through a database
- Streaming data will be provided in the last week of the PoC
- What-if scenarios shared need to be tested on how the system reacts

02 How to pose queries

- Queries may be submitted by email to the address below
- Clarifications will be provided to all bidders in the common shared folder

dtinfo@jnport.gov.in

03 How the data is submitted

- Submission is through Next Cloud
- Each bidder has a folder where the required files are submitted
- The bidder shares access with JNPA

DATA NOTICE

The data shared includes simulated data — **irregularities and anomalies are present by design.** Detecting, documenting and handling them forms part of the evaluation. Data is for PoC use only and is not a validated record of port operations.

DEADLINE

Submission closes **Friday, 07 August 2026 · 5:00 PM**